



Qualification Pack



Fundamentals of CNC Milling Machine Programming

QP Code: SVO/CSC/Q0401

Version: 1.0

NSQF Level: 4

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SVO/CSC/Q0401: Fundamentals of CNC Milling Machine Programming

Brief Job Description

This NOS unit covers identifying NC/CNC milling machine program elements, writing and trialling a machine program for CNC milling machine, and preparing an operation sheet.

Personal Attributes

CNC Milling Machine Programming

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [SVO/CSC/N0401: Fundamentals of CNC Milling Machine Programming](#)

Qualification Pack (QP) Parameters

Sector	Capital Goods
Sub-Sector	Manufacturing
Occupation	Fundamentals of CNC Milling Machine Programming
Country	India
NSQF Level	4
Credits	4
Aligned to NCO/ISCO/ISIC Code	NCO-2015/7223.60
Minimum Educational Qualification & Experience	10th Class + I.T.I with NA of experience OR 12th Class with NA of experience OR Completed 2nd year of the 3-year diploma after 10 with NA of experience
Minimum Level of Education for Training in School	10th Class



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Pre-Requisite License or Training	NA
Minimum Job Entry Age	15 Years
Last Reviewed On	NA
Next Review Date	30/05/2027
NSQC Approval Date	30/05/2024
Version	1.0
Reference code on NQR	G-04-CG-02644-2024-V1-SCTEVTO
NQR Version	1



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SVO/CSC/N0401: Fundamentals of CNC Milling Machine Programming

Description

This NOS unit covers identifying NC/CNC milling machine program elements, writing and trialling a machine program for CNC milling machine, and preparing an operation sheet.

Scope

The scope covers the following :

- Preparatory activities for programming
- Write CNC milling machine program
- Testing and trial run of program

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** follow standard operating procedures (SOPs)
- PC2.**
 - read, interpret and follow the information on written job instructions, specifications, SOPs, part information, CNC milling process plan/sheet, charts, lists, drawings and other applicable reference
 - documents to obtain job requirements, tasks and relevant information
- PC3.**
 - determine the machining operations and processing sequence to be performed on the CNC milling
 - machine through the available documents and information
- PC4.**
 - interpret information such as cutting parameters, revolutions per minute (RPM), feed speed (mm/min), axial depth of cut, and radial depth of cut, etc. from the sketches, job sheets, process
 - plan/sheet etc.
- PC5.**
 - identify the programming method, dimensions, material requirements, cutting tool type, clamping
 - and positioning method, workholding device, tool requirements from the tooling layout, cutting process etc. from the available documents and information and assess their suitability
- PC6.** check the process sheet and match it with the received drawings and other specifications
- PC7.** clarify relevant specifications and procedures from relevant person
- PC8.** select appropriate CNC program elements for milling machine controller
- PC9.**
 - coordinate with the relevant personnel to report and rectify incorrect and inconsistent information in
 - job specification documents
- PC10.** interpret engineering drawings to define milling machine function and tool path geometry
- PC11.** calculate coordinates for tool path or milling functions, as required
- PC12.** write program in standard code format in accordance with SOPs



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- PC13.**
- write the CNC program with commands for tool motions, spindle motions, miscellaneous functions
 - and tool change, in syntax corresponding to the milling machine and control system on which the
 - component will be machined
- PC14.**
- define the programming coordinate system, fully consider factors such as characteristics of the
 - processed material, cutting characteristics of the milling tool, and material condition of the part that
 - need to be removed in the program
- PC15.**
- define processing method, process parameters (including margin, feed speed and processing tool
 - path span, etc.) and auxiliary attributes (including tool setting point, safety surface, milling machine
 - tool attributes, etc.) in the program
- PC16.**
- follow the appropriate method for creating the CNC program, such as text editor or Computer-Aided
 - Manufacturing (CAM) software or controllers on machine
- PC17.**
- ensure the part program is efficient and results in minimal cycle time, with optimal cutting
 - parameters and no unnecessary tool motions
- PC18.** select advanced operations using canned cycles and sub-routines and apply
- PC19.**
- transfer the program to the milling machine by entering it in the console or transmitting it through a
 - wired link or a data transfer device
- PC20.** use the part program in single block mode and check the condition of the tool after each operation
- PC21.** check the sequence of the program as per the process sheet
- PC22.**
- check the annotation information, selection of the processing method, tool size information, tool
 - compensation information etc. in the CNC milling processing program is correct
- PC23.** check for over-cutting, under-cutting or collision interference in the CNC milling program
- PC24.**
- check the CNC milling program, spindle speed and feed speed match the current CNC milling machine
 - tool
- PC25.**
- follow the recommended procedures for calling up the program and dealing with any error messages
 - or faults
- PC26.**
- follow the approved procedures to resolve problems encountered with the programming, loading and
 - editing activities
- PC27.** use the appropriate storage medium such as a computer hard disk for saving the proven program
- PC28.**
- produce operation sheets to specification in accordance with SOPs which includes appropriate, where
 - required
- PC29.**
- select and prepare the appropriate tools and equipment for testing and proving the program,
 - ensuring they are not worn-out or damaged
- PC30.** ensure all the measuring equipment are calibrated and approved for use



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- PC31.** pre-set the tooling appropriately using setting jigs/fixtures
- PC32.** mount tools in the correct positions in the tool turret or magazine
- PC33.** check the tools are mounted in positions corresponding to the tool numbers in the part program
- PC34.** operate milling machine in manual mode to test and prove program
- PC35.** cut a trial part using single block run, dry run and feed, and speed override controls
- PC36.** edit program, if necessary to adjust operation, as required
- PC37.** follow the recommended practices to resolve error messages and faults on the program or equipment
- PC38.** check components to conform to specification
- PC39.**
 - ensure the trial part conforms to drawing specifications in terms of dimensions, surface finishes and
 - geometrical parameters such as concentricity, parallelism, runout, etc.
- PC40.** prepare and verify all technical documents for CNC programs

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** organisation procedures for health, safety and security, individual role and responsibilities in this context
- KU2.** elements of a NC/CNC program
- KU3.** function of elements in controlling the operation of a NC/ CNC milling machine
- KU4.** machining operations to be performed in the manufacture of the given part or product
- KU5.** appropriate type(s) of NC/CNC milling machine to perform the required milling operations
- KU6.** milling operations to be controlled by the program to be written
- KU7.** tool path(s) to be followed when producing the part or product
- KU8.** sequence of milling operations to be programmed
- KU9.** reasons for selecting the chosen tool path(s) and sequence of operations
- KU10.** canned cycles and sub-routines accessible in the particular NC/CNC milling machine
- KU11.** application of each canned cycle and sub-routine available
- KU12.** canned cycles and/or sub-routines to be used in the NC/CNC milling program
- KU13.** reasons for selecting the chosen canned cycles and/or sub-routines
- KU14.** standard codes i.e. D, F, G, M, N, S and T used in the writing of NC/CNC programs
- KU15.** applications of standard codes in NC/CNC programming
- KU16.** procedures for writing NC/CNC programs in standard code format
- KU17.** procedures for completing NC/CNC operation sheets
- KU18.** information to be included in NC/CNC operation sheets
- KU19.** procedures for manual operation of the NC/CNC milling machine
- KU20.** reasons for testing and proving the NC/CNC program
- KU21.** procedures for editing the NC/CNC program via the milling machine controller



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- KU22.** effects of editing on the operation of the NC/CNC milling machine and the part or product to be produced
- KU23.** specifications of the part or product
- KU24.** measuring equipment/techniques to be used to check for conformance with specifications
- KU25.**
 - hazards and control measures associated with numerical and computer-controlled milling machines,
 - including housekeeping
- KU26.** safe work practices and procedures

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** follow instructions, guidelines, procedures, rules, and service level agreements
- GS2.**
 - read, interpret and follow information on written job instructions, specifications, standard operating
 - procedures, charts, lists, drawings and other applicable reference documents
- GS3.** planning and sequencing of operations
- GS4.** checking and clarifying task related information
- GS5.** calculating coordinates of all relevant points on the part or product to be produced
- GS6.**
 - writing NC/CNC program in standard code format and incorporating, where appropriate, canned cycles
 - and sub-routines
- GS7.** producing NC/CNC operation sheet(s)
- GS8.** operating NC/CNC machine in manual mode



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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	35	55	-	20
PC1. follow standard operating procedures (SOPs)	1	-	-	-
PC2. • read, interpret and follow the information on written job instructions, specifications, SOPs, part information, CNC milling process plan/sheet, charts, lists, drawings and other applicable reference • documents to obtain job requirements, tasks and relevant information	1	1	-	1
PC3. • determine the machining operations and processing sequence to be performed on the CNC milling • machine through the available documents and information	1	2	-	1
PC4. • interpret information such as cutting parameters, revolutions per minute (RPM), feed speed (mm/min), axial depth of cut, and radial depth of cut, etc. from the sketches, job sheets, process plan/sheet etc.	1	1	-	1
PC5. • identify the programming method, dimensions, material requirements, cutting tool type, clamping and positioning method, workholding device, tool requirements from the tooling layout, cutting process etc. from the available documents and information and assess their suitability	1	2	-	1
PC6. check the process sheet and match it with the received drawings and other specifications	1	1	-	-
PC7. clarify relevant specifications and procedures from relevant person	-	1	-	-
PC8. select appropriate CNC program elements for milling machine controller	1	1	-	1



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC9. • coordinate with the relevant personnel to report and rectify incorrect and inconsistent information in • job specification documents	-	1	-	-
PC10. interpret engineering drawings to define milling machine function and tool path geometry	1	1	-	1
PC11. calculate coordinates for tool path or milling functions, as required	1	1	-	-
PC12. write program in standard code format in accordance with SOPs	1	1	-	1
PC13. • write the CNC program with commands for tool motions, spindle motions, miscellaneous functions • and tool change, in syntax corresponding to the milling machine and control system on which the • component will be machined	2	4	-	1
PC14. • define the programming coordinate system, fully consider factors such as characteristics of the • processed material, cutting characteristics of the milling tool, and material condition of the part that • need to be removed in the program	2	3	-	1
PC15. • define processing method, process parameters (including margin, feed speed and processing tool • path span, etc.) and auxiliary attributes (including tool setting point, safety surface, milling machine • tool attributes, etc.) in the program	1	2	-	-
PC16. • follow the appropriate method for creating the CNC program, such as text editor or Computer-Aided • Manufacturing (CAM) software or controllers on machine	1	2	-	1
PC17. • ensure the part program is efficient and results in minimal cycle time, with optimal cutting • parameters and no unnecessary tool motions	1	1	-	-
PC18. select advanced operations using canned cycles and sub-routines and apply	1	1	-	1



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC19. transfer the program to the milling machine by entering it in the console or transmitting it through a wired link or a data transfer device	1	1	-	-
PC20. use the part program in single block mode and check the condition of the tool after each operation	1	2	-	1
PC21. check the sequence of the program as per the process sheet	1	1	-	-
PC22. check the annotation information, selection of the processing method, tool size information, tool compensation information etc. in the CNC milling processing program is correct	1	2	-	1
PC23. check for over-cutting, under-cutting or collision interference in the CNC milling program	1	1	-	-
PC24. - check the CNC milling program, spindle speed and feed speed match the current CNC milling machine - tool	1	2	-	1
PC25. - follow the recommended procedures for calling up the program and dealing with any error messages - or faults	1	1	-	-
PC26. follow the approved procedures to resolve problems encountered with the programming, loading and editing activities	-	1	-	-
PC27. use the appropriate storage medium such as a computer hard disk for saving the proven program	-	1	-	-
PC28. - produce operation sheets to specification in accordance with SOPs which includes appropriate, where - required	1	2	-	1



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC29. - select and prepare the appropriate tools and equipment for testing and proving the program, - ensuring they are not worn-out or damaged	1	2	-	1
PC30. ensure all the measuring equipment are calibrated and approved for use	-	1	-	-
PC31. pre-set the tooling appropriately using setting jigs/fixtures	1	1	-	1
PC32. mount tools in the correct positions in the tool turret or magazine	1	1	-	-
PC33. check the tools are mounted in positions corresponding to the tool numbers in the part program	1	1	-	-
PC34. operate milling machine in manual mode to test and prove program	1	2	-	1
PC35. cut a trial part using single block run, dry run and feed, and speed override controls	1	2	-	1
PC36. edit program, if necessary to adjust operation, as required	1	1	-	-
PC37. follow the recommended practices to resolve error messages and faults on the program or equipment	-	1	-	-
PC38. check components to conform to specification	1	1	-	1
PC39. - ensure the trial part conforms to drawing specifications in terms of dimensions, surface finishes and - geometrical parameters such as concentricity, parallelism, runout, etc.	-	1	-	-
PC40. prepare and verify all technical documents for CNC programs	1	1	-	-
NOS Total	35	55	-	20



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National Occupational Standards (NOS) Parameters

NOS Code	SVO/CSC/N0401
NOS Name	Fundamentals of CNC Milling Machine Programming
Sector	Capital Goods
Sub-Sector	
Occupation	Fundamentals of CNC Milling Machine Programming, Fundamentals of CNC Milling Machine Programming
NSQF Level	4
Credits	4
Version	1.0
Last Reviewed Date	30/05/2024
Next Review Date	30/05/2027
NSQC Clearance Date	30/05/2024

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1.

Assessment System Overview:

-

Batches assigned to the assessment agencies for conducting the assessment on SDMS/SIP or email

-

Assessment agencies send the assessment confirmation to TP

-

Assessment agency deploys the ToA certified Assessor for executing the assessment



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•

Training agency monitors the assessment process & records

2.

Testing Environment:

•

Confirm that the centre is available at the same address as mentioned on SDMS or SIP

•

Check the duration of the training.

•

Check the Assessment Start and End time to be as 10 a.m. and 5 p.m.

•

If the batch size is more than 30, then there should be 2 Assessors.

•

Check that the allotted time to the candidates to complete Theory & Practical Assessment is correct.

•

Check the mode of assessment—Online (TAB/Computer) or Offline (OMR/PP).

•

Confirm the number of TABs on the ground are correct to execute the Assessment smoothly.

•

Check the availability of the Lab Equipment for the particular Job Role.

3.

Assessment Quality Assurance levels / Framework:

•

Question papers created by the Subject Matter Experts (SME)

•

Question papers created by the SME verified by the other subject Matter Experts



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-

Questions are mapped with NOS and PC

-

Question papers are prepared considering that level 1 to 3 are for the unskilled & semi-skilled individuals, and level 4 and above are for the skilled, supervisor & higher management

-

Assessor must be ToA certified & trainer must be ToT Certified

-

Assessment agency must follow the assessment guidelines to conduct the assessment

4.

Types of evidence or evidence-gathering protocol:

-

Time-stamped & geo tagged reporting of the assessor from assessment location

-

Centre photographs with signboards and scheme specific branding

-

Biometric or manual attendance sheet (stamped by TP) of the trainees during the training period

-

Time-stamped & geo tagged assessment (Theory + Viva + Practical) photographs & videos

5.

Method of verification or validation:

-

Surprise visit to the assessment location

-

Random audit of the batch

-



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Random audit of any candidate

6.

Method for assessment documentation, archiving, and access

-

Hard copies of the documents are stored

-

Soft copies of the documents & photographs of the assessment are uploaded / accessed from Cloud Storage

-

Soft copies of the documents & photographs of the assessment are stored in the Hard Drives

Minimum Aggregate Passing % at QP Level : 70

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
SVO/CSC/N0401.Fundamentals of CNC Milling Machine Programming	35	55	0	20	110	100
Total	35	55	-	20	110	100



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Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training



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Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.



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Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.